# FBMSNet: A Filter-Bank Multi-Scale Convolutional Neural Network for EEG-Based Motor Imagery Decoding

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***Abstract*—*Object:* Motor imagery (MI) is a mental process widely utilized as the experimental paradigm for brain-computer interfaces (BCIs) across a broad range of basic science and clinical studies. However, decoding intentions from MI remains challenging due to the inherent complexity of brain patterns relative to the small sample size available for machine learning. *Approach:* This paper proposes an end-to-end Filter-Bank Multiscale Convolutional Neural Network (FBMSNet) for MI classification. A filter bank is first employed to derive a multiview spectral representation of the EEG data. Mixed depthwise convolution is then applied to extract temporal features at multiple scales, followed by spatial filtering to mitigate volume conduction. Finally, with the joint supervision of cross-entropy and center loss, FBMSNet obtains features that maximize interclass dispersion and intraclass compactness. *Main results:* We compare FBMSNet with several state-of-the-art EEG decoding methods on two MI datasets: the BCI Competition IV 2a dataset and the OpenBMI dataset. FBMSNet significantly outperforms the benchmark methods by achieving 79.17% and 70.05% for four-class and two-class hold-out classification accuracy, respectively. *Significance:* These results demonstrate the efficacy of FBMSNet in improving EEG decoding performance toward more robust BCI applications. The FBMSNet source code is available at https://github.com/Want2Vanish/FBMSNet.**

***Index Terms*—Brain computer interface (BCI), electroencephalography (EEG), motor imagery, convolutional neural network, mixed depthwise convolution.**

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## I. Introduction

Motor imagery (MI), is the mental rehearsal of movement execution without any physical movement [1]. Due to the advantage of not requiring an external stimulus, MI has attracted considerable attention for building electroencephalography (EEG)-based brain-computer interface (BCI) systems. The oscillations in the motor area elicited by MI are known as sensorimotor rhythms (SMRs). It is well known that different classes of MI are associated with distinct spectrospatial distributions of SMRs. By decoding the SMRs, one can recognize user movement intent and help users control physical or virtual devices, thus improving the ability of humans to communicate and interact with the world [2]. For example, MI has been successfully applied to stroke recovery [3], [4], exoskeleton robot arm control [5], [6] and wheelchair control [7].

However, decoding motor intentions is nontrivial, largely due to several characteristics of EEG signals, i.e., low signal-to-noise ratio, sensitivity to noise, and high intertrial variability. These issues are aggravated by the scarcity of data available for machine learning. Existing studies on MI classification can be grouped into two approaches: classical machine learning approaches, and deep learning based approaches. Classical MI decoding methods first compute predefined features, and then user intents are recognized using established pattern classifiers, e.g., linear discriminant analysis (LDA) and support vector machines (SVMs). Spatial filtering followed by either band-power or time-point feature extraction is commonly used in existing EEG-based BCIs. In particular, common spatial patterns (CSPs) and the variant filter bank CSP (FBCSP) [8] are representative spatial-filtering-based methods and are widely employed for MI decoding. However, these classical approaches are susceptible to intertrial variability in EEG data and are highly dependent on handcrafted features; hence, the resulting models may be poorly generalized to unseen data.

In recent years, a number of deep learning-based MI-decoding methods have been proposed and have achieved excellent classification performance. Compared with traditional handcrafted-feature-based decoding methods, deep learning-based methods are able to automatically extract features from EEG signals in a hierarchical manner. Accordingly, the feature-representation capability of these methods is more powerful due to their deep nonlinear network architectures. For example, Schirrmeister et al. [9] proposed deep and shallow convolutional neural networks (CNNs), i.e., Deep ConvNet and Shallow ConvNet, to process EEG data. Both networks perform simultaneous feature extraction and classification in an end-to-end fashion with the utilization of CNNs. Lawhern et al. [10] proposed EEGNet, a compact deep learning framework that used three convolutional layers to extract temporal and spatial patterns from EEG data. These CNN models outperform conventional feature-based MI decoding methods, such as FBCSP-SVM.

Nonetheless, the abovementioned CNN models employ a fixed convolutional kernel size along the time dimension to extract temporal information. As suggested in [11], the optimal kernel size may vary from subject to subject or session to session. Hence, to improve decoding accuracy, the study in [11] proposed a CNN with hybrid convolution scales. Similarly, the study in [12] designed a multibranch 3D-CNN combining a new 3D EEG representation, with excellent robustness on multiple subjects. More recently, Zhang et al. developed the EEG-Inception neural network, which employs multiple (three or five in [13]) 1-D convolutional layers with different kernel sizes to extract multiscale features, achieving an average accuracy of 88.4% and 88.6% on the 2008 BCI Competition IV 2a (four-class) and 2b (binary) datasets, respectively. Therefore, the multiscale information extracted by a multibranch structure may enhance the MI decoding performance. However, the use of inception blocks or more complex multibranch structures also significantly increases the number of trainable parameters of neural networks. Hence, more training data are necessary to avoid overfitting. Moreover, more computation time and memory are needed.

Recent studies have shown that using spectrally-filtered EEG data as the network input is beneficial, especially for small training datasets. For instance, Robinson et al. [14] proposed a multiview EEG representation that preserves temporal, spatial and spectral information, and reported excellent classification performance. Mane et al. adopted the same multiview representation and proposed a novel variance layer that achieved state-of-the-art performance on the BCIC-IV 2a and OpenBMI datasets [15]. However, these multiview methods suffer from the same limitations as in conventional CNN models due to the lack of multiscale feature extraction in the time domain, impeding their ability to learn discriminative spectral–spatial feature representations on an individual basis.

Due to its simplicity and effectiveness, the cross-entropy (CE) loss is widely employed as a cost function to optimize the parameters of deep neural networks. However, recent studies [16]–[18] suggest that conventional CE is ineffective in reducing intraclass variation, which is aggravated by the nonstationarity of EEG signals [19], [20]. To address this issue, in addition to the CE loss, many methods have employed other loss functions, such as the triplet loss [21] and center loss [18], [22], as auxiliary costs to learn models that can decrease the between-class similarity and within-class diversity. However, preparing sample pairs is time-consuming for the triplet and contrastive loss. Hence, Wen et al. [22] proposed the center loss to address this problem. The center loss learns a center for the deep representations of every class and minimizes the distances between the learned representations and their corresponding class centers. Combining the CE loss and the center loss, deep neural network models are able to learn separable and discriminative representations [22].

Inspired by previous pioneering studies, in this work, we propose a new CNN model, named the Filter-Bank Multi-Scale Convolutional Network (FBMSNet), for MI decoding. FBMSNet is an efficient and lightweight multiscale feature extraction CNN architecture for MI classification. Specifically, to achieve multiscale feature learning, we adopt mixed depthwise convolution (MixConv) [23], which is able to reduce the dependency of training data and speed up the classification process. Furthermore, to distinguish similar categories in a better way and decrease the influence of interclass dispersion and within-class variance, we not only minimize the CE loss function but also introduce the center loss function [22]. With this joint supervision, FBMSNet is capable of learning deep features with two key learning objectives as much as possible, inter-class separability and intra-class compactness as much as possible, which are crucial to MI recognition. The performance of FBMSNet was validated on the BCI Competition IV 2a (BCIC-IV-2a) dataset and the OpenBMI dataset. The classification results indicated that FBMSNet significantly outperformed the baseline methods on both datasets under the cross-validation and hold-out analyses.

The main contributions of this work can be summarized as follows

1) We propose FBMSNet, a novel multiscale temporal convolutional neural network, for MI decoding tasks. FBMSNet employs MixConv to extract multiscale temporal features. With the joint supervision of the CE loss and center loss, the learned feature representations can not only enhance the intra-class compactness, but also improve the inter-class separability.
2) Extensive ablation studies and numerical experiments are conducted, suggesting the superior performance FBMSNet over FBCSP+SVM, Shallow ConvNet, Deep ConvNet, EEGNet, and FBCNet.

The rest of this paper is organized as follows. Section II describes the structure and optimization of FBMSNet. The experimental protocol and results are presented in Section III and Section IV, respectively. Section V provides a discussion, followed by a brief conclusion in Section VI.

## II. Methods

In this section, we first describe the multiview representation of EEG data, which is employed as the input to the proposed FBMSNet. Then, we present the specific network architecture of FBMSNet. Finally, the cost function for optimizing the unknown parameters of FBMSNet is introduced.

### A. EEG Representation

Since different MI classes may differ in their corresponding spectro-spatial SMR patterns, FBMSNet employs a multiview representation of the EEG data as the input in which each view represents a narrow-band localized EEG. Specifically, let $\boldsymbol{X} \in \mathbb{R}^{C \times T}$ denote a single-trial raw EEG data and let $\boldsymbol{y} \in \{0, 1, ..., N_c - 1\}$ denote its corresponding label, where $C$ represents the number of EEG channels, $T$ represents the

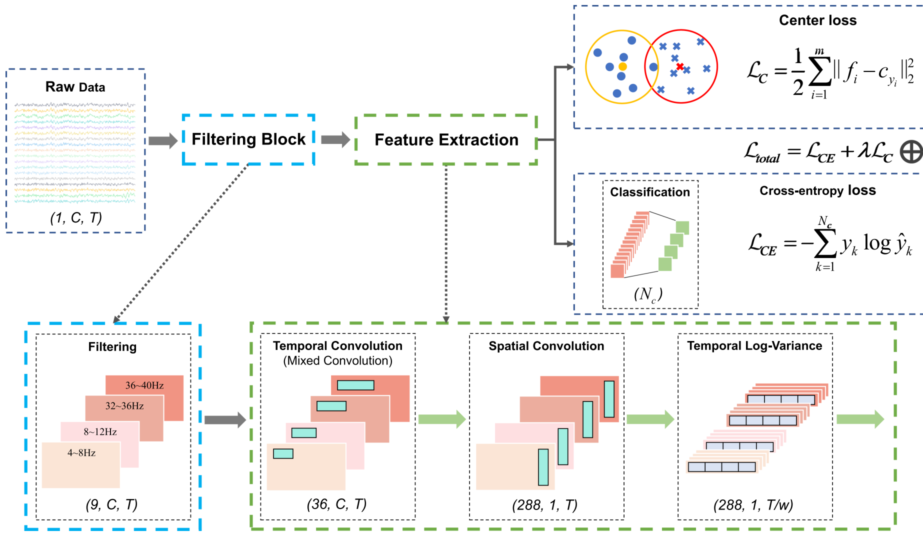


Fig. 1. Structure of FBMSNet. $C$, $T$ and $N_b$ denote the number of EEG channels, the number of time points and the number of frequency bands, respectively. $w$ is the window length of the variance layer, and $N_c$ is number of output classes.

TABLE I
PARAMETERS OF FBMSNET

| Layer | Filter | Size | Stride | Activation | Options | Output |
|---|---|---|---|---|---|---|
| Input | | | | | | (1, C, T) |
| Filtering | | (1, C, T) | | | | ($N_b$, C, T) |
| Conv2D | 9 | (1, 15) | 1 | Linear | padding = same | (9, C, T) |
| Conv2D | 9 | (1, 31) | 1 | Linear | padding = same | (9, C, T) |
| Conv2D | 9 | (1, 63) | 1 | Linear | padding = same | (9, C, T) |
| Conv2D | 9 | (1, 125) | 1 | Linear | padding = same | (9, C, T) |
| Concatenate | | | | | dim = 1 | (36, C, T) |
| BatchNorm | | | | | | (36, C, T) |
| DepthwiseConv2D | 288 | (C, 1) | 1 | Linear | group = 36 | (288, 1, T) |
| BatchNorm | | | | | | (288, 1, T) |
| Activation | | | | swish | | (288, 1, T) |
| Reshape | | | | | | (288, T // w, w) |
| Variance | | (w) | | | | (288, T // w, 1) |
| Flatten | | | | | | (288 * (T // w)) |
| FC | $N_c$ | | | softmax | | ($N_c$) |

$C$ is number of EEG channels, $T$ is number of time points, $N_b$ is the number of frequency bands, $w$ is the window length of variance layer, and $N_c$ is number of output classes.

sampled time points, and $N_c$ is the total number of distinct classes. The multiview representation of the EEG data, $\boldsymbol{X}_{\text{FB}} \in \mathbb{R}^{N_b \times C \times T}$, where $N_b$ represents the number of frequency bands, is generated by spectrally filtering the raw EEG $\boldsymbol{X}$ with a filter bank $\boldsymbol{F}$ consisting of a set of narrow-band filters. In this study, inspired by FBCSP [8], the filter bank $\boldsymbol{F}$ is constructed using 9 nonoverlapping frequency bands, each with a 4 Hz bandwidth and spanning from 4 to 40 Hz (i.e., 4–8, 8–12..., 36–40 Hz). Filtering in each band is performed using a Chebyshev Type II filter with a transition bandwidth of 2 Hz and stop-band ripple of 30 dB.

## B. Architecture of FBMSNet

FBMSNet consists of four blocks, (1) a temporal convolution block, (2) a spatial convolution block, (3) a temporal log-variance block, and (4) a fully connected layer for classification. The first block is designed to learn the multiscale temporal information from the multiview EEG representations, and the second block aims to learn the spatial information from each temporal feature map. Subsequently, the third block computes the temporal variance of each time series. Finally, all representations are flattened and fed to the fully connected layer with softmax as the activation function. An overview of FBMSNet is depicted in Fig. 1, and the detailed configuration of the network architecture is provided in Table I, with each block described below.

1) Temporal convolution block. To overcome the limitations of previous multiscale feature extraction methods based on multibranch or inception blocks, we employ a mixed depthwise convolution (MixConv) [23] to extract multiscale temporal information from the multiview representation of EEG data. Specifically, this block splits the feature map into multiple groups by views, with each group using kernels of different sizes for convolution, and then combining them. Compared to prior multiscale

methods, this reduces the number of parameters and mitigates the risk of overfitting. As in TSception [24], we set the length of the temporal kernel to a specific ratio of the EEG sampling rate, with the ratio coefficients being $[0.5, 0.25, 0.125, 0.0625]$. Accordingly, the inputs were divided equally into 4 groups, each of which learns 9 time-domain filters.

2) Spatial convolution block. This block starts with depthwise convolution, followed by a batch normalization layer and Swish nonlinearity activation function. The size of the convolution kernel is set to $(C, 1)$ to span all channels, effectively acting as a spatial filter. In addition, we regularize each convolution kernel using a maximum norm constraint of 2 on its weights, i.e., $\|\boldsymbol{w}\|^2 < 2$ (weight-normalisation).

3) Temporal log-variance block. After spatiotemporal filtering in the preceding layers, MI data still contain a large number of time points, and it is necessary to reduce the temporal dimensionality by efficiently extracting the most relevant features. For SMRs, computing the variance proves to be more effective than maximum pooling and average pooling [15]. Therefore, this block computes the log-variance of each time series in a nonoverlapping window of size $\boldsymbol{w}$. The log-transform is used to enhance the Gaussianity of the variance feature.

4) Classification. The log-variance features are then provided to a fully connected (FC) layer with linear activation. The output of the FC is then passed to the softmax layer to yield the output probability of each class. The FC layer weights are also regularized using a maximum norm constraint of 0.5, i.e., $\|\boldsymbol{w}\|^2 < 0.5$ (weight-normalization).

### C. Optimization Functions of FBMSNet

Cross entropy (CE) is a widely used loss function for optimizing classification models, and is defined as

$$\mathcal{L}_{CE} = -\sum_{k=1}^{N_c} y_k \log \hat{y}_k, \tag{1}$$

where $y$ is the true label and $\hat{y}$ is the predicted label. $N_c$ denotes the number of classes.

Nonetheless, as noted in [22], CE loss is susceptible to intraclass variation. To remedy this issue, in addition to the CE loss, we introduce an additional loss function, i.e., center loss, which enables us to minimize the intraclass variability. The center loss is defined as

$$\mathcal{L}_C = \frac{1}{2}\sum_{i=1}^{m} \|\boldsymbol{f}_i - \boldsymbol{c}_{y_i}\|_2^2, \tag{2}$$

where $\boldsymbol{f}_i \in \mathbb{R}^{d\times 1}$ is the feature vector of sample $i$, and $\boldsymbol{c}_{y_i} \in \mathbb{R}^{d\times 1}$ is the feature center of the class to which sample $i$ belongs. $m$ is the batch size of the mini-batch. Following the suggestion in [22], instead of updating the feature centers with respect to the entire training set, we update the feature centers $\boldsymbol{c}_k (k = 1, \cdots, N_c)$ based on the mini-batch in each iteration.

**Algorithm 1:** Optimization Procedure of FBMSNet.

**Input:** $N$ trials of EEG training data $\boldsymbol{X} \in \mathbb{R}^{N\times C\times T}$, ground truth labels $\boldsymbol{Y} \in \mathbb{R}^{N\times 1}$, initialized parameters of FBMSNet $\boldsymbol{\Theta}$, hyper-parameters $\lambda$, $\alpha$ and learning rate $\mu$.

**Output:** the parameters of FBMSNet $\boldsymbol{\Theta}$

**for** *each training iteration* **do**
- `// Fetch one batch of training data`
- $(\boldsymbol{x}, \boldsymbol{y}) \in (\boldsymbol{X}, \boldsymbol{Y})$;
- `// Forward-propagation`
- $\hat{\boldsymbol{y}},\ \boldsymbol{f} = \text{FBMSNet}_{\boldsymbol{\Theta}}(\boldsymbol{x})$;
- `// Compute the total loss`
- $\mathcal{L} = \mathcal{L}_{CE}(\hat{\boldsymbol{y}}, \boldsymbol{y}) + \lambda \cdot \mathcal{L}_C(\boldsymbol{f}, \boldsymbol{y})$;
- `// Back-propagation`
- $\frac{\partial \mathcal{L}}{\partial \boldsymbol{f}} = \frac{\partial \mathcal{L}_{CE}}{\partial \boldsymbol{f}} + \lambda \cdot \frac{\partial \mathcal{L}_C}{\partial \boldsymbol{f}}$;
- `// Update the` $c_k$ `for each class`
- $\boldsymbol{c}_k \leftarrow \boldsymbol{c}_k - \alpha \Delta \boldsymbol{c}_k$;
- `// Update the parameters` $\boldsymbol{\Theta}$
- $\boldsymbol{\Theta} \leftarrow \boldsymbol{\Theta} - \mu \sum_i^m \frac{\partial \mathcal{L}}{\partial \boldsymbol{f}} \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{\Theta}}$;

**end**

Specifically, the feature center of the $k$-th class, $\boldsymbol{c}_k$, is updated as

$$\boldsymbol{c}_k \leftarrow \boldsymbol{c}_k - \alpha \Delta \boldsymbol{c}_k$$
$$\Delta \boldsymbol{c}_k = \frac{\sum_{i=1}^{m} \delta(i,k)(\boldsymbol{c}_k - \boldsymbol{f}_i)}{1 + \sum_{i=1}^{m} \delta(i,k)}, \tag{3}$$

where $\delta(i, k) = 1$ if sample $i$ belongs to the $k$-th class, and $\delta(i, k) = 0$ otherwise. To avoid large perturbations due to a few mislabeled samples, we use a scalar $\alpha$ to control the learning rate of the centers. Given that the performance of the center loss remains largely stable across a wide range of values for $\alpha$ [22], we empirically set the value of $\alpha$ to be 0.01.

Combining the center loss and CE loss is advantageous in that the former enhances the intraclass compactness of features, while the latter promotes the interclass separability of features. Consequently, we obtain the following loss $\mathcal{L}_{\text{total}}$ for optimizing the parameters of FBMSNet:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CE} + \lambda \mathcal{L}_C, \tag{4}$$

where $\lambda$ is the trade-off scalar to balance the two losses, which is empirically set to 0.0005 in this work. The impact of $\lambda$ and $\alpha$ on MI decoding performance is discussed in Section V.

The detailed optimization procedure of FBMSNet is shown in Algorithm 1. The source code of the proposed method is available at https://github.com/Want2Vanish/FBMSNet.

## III. Experiments

### A. Data Description

An overview of the datasets for evaluating FBMSNet and baseline methods is provided as follows:

- *BCIC-IV-2a [25]:* The BCIC-IV-2a dataset consists of EEG data from 9 subjects collected over 2 sessions. A total of 4 MI classes are included: MI of the left hand, right hand, both feet, and tongue. The EEG data were recorded using 22 electrodes with a sampling frequency of 250 Hz. There were 72 trials of every class in each session and every trial was 4s in duration. In our analysis, all 22 channels and the entire 4-second data were used.

- *OpenBMI [26]:* This dataset contains EEG signals of 62 channels for 2-class MI tasks (i.e., executing the imagination of left and right hand) recorded from 54 right-handed healthy subjects (29 males and 25 females, aged 24–35), among whom 38 subjects were naive BCI users. The others had prior experience participating in BCI experiments. Following [15], we selected 20 channels in the motor region (FC-5/3/1/2/4/6, C-5/3/1/z/2/4/6, and CP-5/3/1/z/2/4/6) in our analysis.

### B. Benchmark Methods

An overview of the benchmark methods is described as follows:

- *FBCSP-SVM [8]:* FBCSP first uses a set of bandpass filters in conjunction with the CSP algorithm to extract the optimal spatial features from the subject-specific frequency band and then trains an SVM for classification. Of note, FBCSP was the best performing method for the BCI Competition IV 2a dataset during the competition.
- *Deep ConvNet [9]:* Deep ConvNet has four convolution-max-pooling blocks, with a specialized first block designed to handle the EEG input, followed by three standard convolution-max pooling blocks and a dense softmax classification layer.
- *Shallow ConvNet [9]:* Shallow ConvNet is specifically designed to decode band power features. It has two layers for temporal convolution and spatial filtering, respectively. These steps are similar to the bandpass filtering and CSP analysis of FBCSP.
- *EEGNet [10]:* EEGNet starts with a temporal convolution to learn frequency filters and then uses a depthwise convolution to learn frequency-specific spatial filters. The separable convolution is a combination of a depthwise convolution, followed by a pointwise convolution, which learns how to optimally mix the feature maps together.
- *FBCNet [15]:* FBCNet adopts depthwise convolution to extract spectral-spatial features from a multiview EEG representation, followed by a variance layer for feature extraction. FBCNet is the STOA method for the BCI Competition IV 2a dataset.

In this study, we adopted the FBCSP-toolbox[1] as the implementation of FBCSP-SVM and the source code (based on the PyTorch framework) provided by [15] as the implementations of Deep ConvNet, Shallow ConvNet, EEGNet and FBCNet.

One major modification was made to EEGNet due to the difference in the sampling frequencies of the data. The original EEGNet was proposed for data with a 128 Hz sampling frequency. However, both datasets in this work have a sampling frequency of 250 Hz. Therefore, we multiply the lengths of the temporal kernels and temporal pooling layers in their architectures by 2 to approximately correspond to the sampling rate in our model, as is done in [15].

[1]Tushar C. FBCSP Toolbox. https://fbcsptoolbox.github.io/.

### C. Experimental Setups

To evaluate the performance of FBMSNet, cross-validation (CV) and hold-out (HO) analyses were performed. In the CV analysis, both the training and test data were taken from session 1 to avoid the confounding influence of inter-session variability, as done in previous works. Specifically, the complete data from session 1 were evenly split into 10 folds, and each fold was category-balanced. Nine folds were used for training, and the remaining fold was used for testing. This means that each fold is used as a test set once. For fairness, we also ensured that this allocation remained constant throughout the experiments for the different methods. In the HO analysis, to understand the impact of intersession variability on classification performance, the training and test sets were taken from completely different sessions as a way to assess the ability of the model to extract highly general discriminative features. Specifically, the session 1 data were used for training and the trained model was tested on session 2 data. The data split of BCIC-IV-2a for the HO analysis is consistent with previous studies [8]. Notably, we mainly focus on the subject-specific MI decoding, whereas cross-subject validation is not the goal for the current work.

The significance of the classification accuracy was assessed by the Kruskal–Wallis test. If a test statistic from the Kruskal–Wallis test is significant, then Wilcoxon rank sum tests are subsequently performed in FBMSNet against each of the other benchmark methods to determine if FBMSNet yields significantly better performance.

### D. Training Details

To prevent overfitting and reduce the number of epochs needed to train the model, a two-stage training strategy as in [9] was used in this work. Specifically, during the training phase, the training data were split into a training set and a validation set. In the first stage, only the training set was used, and the training was stopped if there was no increase in the validation set accuracy for 200 consecutive epochs. During the second training stage, all training data were employed, and the training was stopped when the average loss function on the validation set fell below the value of the training set at the end of the first training stage. The detailed training procedure is provided in Algorithm 2. In this work, the maximum number of training epochs was limited to 1500 and 600 for training stages 1 and 2, respectively.

To minimize the loss function, in this work, we employed the Adam optimizer with default settings (learning rate $= 0.001$, betas $= 0.9, 0.999$) [27]. In the 10-fold CV analysis, data from one fold of the nine training folds were retained as the validation set. In the HO analysis, 20% of the training data were used as a validation set. During both the CV and HO analyses, the test fold/set never appeared in any of the two-stage training phases.

**Algorithm 2:** Two-stage Training Strategy.

**Input:** $N$ trials of EEG training data $\boldsymbol{X}_{train} \in \mathbb{R}^{N\times C\times T}$, ground truth labels $\boldsymbol{Y}_{train} \in \mathbb{R}^{N\times 1}$, initialized parameters of FBMSNet $\boldsymbol{\Theta}$, the maximum number of training epochs maxEpochs, hyper-parameters $\lambda$, $\alpha$ and learning rate $\mu$.

**Output:** the parameters of FBMSNet $\boldsymbol{\Theta}$

Stage 1:
$\boldsymbol{X}_{sub}, \boldsymbol{X}_{val}$ = func_split($\boldsymbol{X}_{train}$);
$\boldsymbol{Y}_{sub}, \boldsymbol{Y}_{val}$ = func_split($\boldsymbol{Y}_{train}$);
**while** *epoch* < *max_epochs_1st* **do**
  Algorithm 1 ($\boldsymbol{X}_{sub}, \boldsymbol{Y}_{sub}$, $\boldsymbol{\Theta}$, $\lambda$, $\alpha$, $\mu$);
  $pred_{sub}, loss_{sub}$ = func_predict ($\boldsymbol{X}_{sub}, \boldsymbol{Y}_{sub}$, $\boldsymbol{\Theta}$);
  $pred_{val}, loss_{val}$ = func_predict ($\boldsymbol{X}_{val}, \boldsymbol{Y}_{val}$, $\boldsymbol{\Theta}$);
  **if** $loss_{val}$ < *min_loss* **then**
    min_loss = $loss_{val}$;
    best_model = $\boldsymbol{\Theta}$;
  **end**
  doStop = func_stopCheck($loss_{val}$, min_loss);
  **if** *doStop* **then**
    $\epsilon$ = $loss_{sub}$;
    break;
  **end**
  epoch += 1;
**end**
Stage 2:
**while** *epoch* < *max_epochs_2nd* **do**
  Algorithm 1 ($\boldsymbol{X}_{train}, \boldsymbol{Y}_{train}$, $\boldsymbol{\Theta}$, $\lambda$, $\alpha$, $\mu$);
  $pred_{val}, loss_{val}$ = func_predict ($\boldsymbol{X}_{val}, \boldsymbol{Y}_{val}$, $\boldsymbol{\Theta}$);
  **if** $loss_{val}$ < *min_loss* **then**
    min_loss = $loss_{val}$;
    best_model = $\boldsymbol{\Theta}$;
  **end**
  doStop = func_stopCheck($loss_{val}$, $\epsilon$);
  **if** *doStop* **then**
    $\boldsymbol{\Theta}_{best}$ = best_model;
    break;
  **end**
  epoch += 1;
**end**

The parameters for all deep learning models are initialized using randomization. For FBMSNet, the centers of the center loss are initialized by random sampling from a univariate Gaussian distribution with a mean of 0 and variance of 1.

## IV. Results

*Result I: FBMSNet achieved significantly higher classification accuracy than baseline methods*

Table II presents complete classification results for both datasets using FBMSNet and the compared methods. For the CV analysis, FBMSNet obtains an accuracy of 83.83% for the four-class MI decoding on the BCIC-IV-2a dataset, which is significantly higher than the four baseline methods. For the OpenBMI dataset, FBMSNet achieves the largest mean accuracy of 75.23%, although it is not statistically significant compared to EEGNet and FBCNet. For the HO analysis, the superior performance of FBMSNet is more apparent, achieving 79.17% four-class classification accuracy on the BCIC-IV-2a dataset and 70.05% two-class decoding accuracy on the OpenBMI dataset, which is significantly larger than the other four benchmark methods. More detailed classification results are presented in the supplementary document.

TABLE II
Classification Accuracies (mean ± std) in % for CV and HO Schemes on the BCI-IV-2a and OpenBMI Datasets Using of FBMSNet and the Compared Methods

| | Methods | BCIC-IV-2a | OpenBMI |
|---|---|---|---|
| CV | FBCSP-SVM | $75.89 \pm 13.08$ * | $64.61 \pm 16.92$ ** |
| | Shallow ConvNet | $76.35 \pm 12.27$ ** | $67.08 \pm 14.46$ ** |
| | Deep ConvNet | $72.20 \pm 13.53$ ** | $68.33 \pm 12.23$ ** |
| | EEGNet | $73.13 \pm 8.73$ ** | $70.89 \pm 12.89$ ** |
| | FBCNet | $79.03 \pm 11.29$ * | $74.70 \pm 13.82$ |
| | FBMSNet(w/o $\mathcal{L}_C$) | $81.81 \pm 9.96$ * | $\mathbf{75.93 \pm 13.90}$ |
| | FBMSNet | $\mathbf{83.83 \pm 8.47}$ | $75.23 \pm 15.44$ |
| HO | FBCSP-SVM | $68.06 \pm 13.30$ * | $60.36 \pm 14.84$ ** |
| | Shallow ConvNet | $74.50 \pm 12.34$ ** | $61.30 \pm 15.12$ ** |
| | Deep ConvNet | $72.21 \pm 11.43$ * | $60.77 \pm 11.31$ ** |
| | EEGNet | $73.15 \pm 8.04$ * | $63.63 \pm 10.97$ ** |
| | FBCNet | $76.20 \pm 12.42$ * | $67.19 \pm 14.24$ ** |
| | FBMSNet(w/o $\mathcal{L}_C$) | $78.82 \pm 11.36$ | $68.41 \pm 14.16$ * |
| | FBMSNet | $\mathbf{79.17 \pm 11.13}$ | $\mathbf{70.05 \pm 14.51}$ |

Bold highlights the best decoding accuracies. The * and ** indicate that the decoding accuracies of FBMSNet are significantly higher than those of the compared methods with $p < 0.05$ and $p < 0.01$, respectively.

Meanwhile, even when FBMSNet is only trained by CE, it still yields significant improvement compared with EEGNet, Deep Convnet, Shallow Convnet, and FBCNet, indicating that the architecture of FBMSNet can improve the MI decoding performance. By adding the center loss as an auxiliary cost, FBMSNet is able to further improve the decoding accuracy except for the subject-specific cross-validation analysis of the OpenBMI dataset.

Fig. 2 presents the confusion matrix results for the HO schema. For each dataset, the confusion matrix results in a composite of all subjects. As shown Fig. 2, FBMSNet has the largest values along the diagonal and smallest values off the diagonal for all datasets. Additionally, FBCNet is more confusing than FBMSNet regard to identifying MI other than the right hand on the BCI-IV-2a dataset.

To further analyze the performance of FBMSNet, we compare the accuracies of all methods for subjects with the 25% highest and 25% lowest HO accuracy for the OpenBMI dataset. As shown in Table III, for the top 25% of subjects, FBCSP-SVM obtains higher accuracy than Deep ConvNet and EEGNet but has lower accuracy than FBCNet, which is in line with the result of [15]. Moreover, FBMSNet obtains even better accuracy than FBCNet. However for the bottom 25% of subjects, the performance of FBCSP-SVM is worse than that of deep learning methods, and FBMSNet achieves the largest average accuracy among the five MI decoding methods.

*Result II: FBMSNet was least affected by datasets with fewer training samples*

It is well known that overfitting usually occurs for deep neural networks with insufficient training data. Here, we test

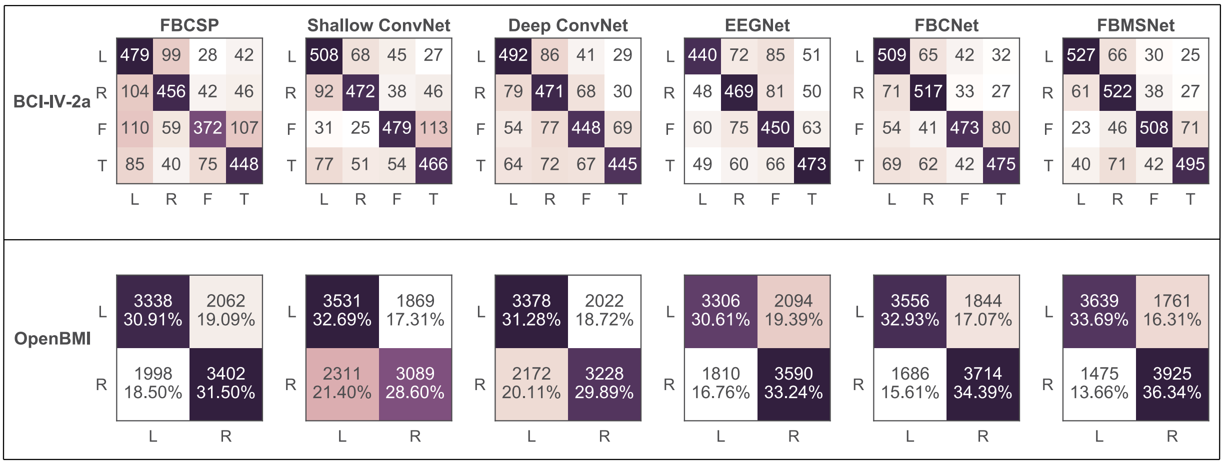


Fig. 2. Confusion matrices for FBMSNet and baseline methods, where each column represents the actual values and each row depicts the predicted values of the model. The results are shown for the BCI-IV-2a and OpenBMI datasets, in the HO schema, where L, R, F, and T refer to MI of left hand, right hand, feet, and tongue, respectively.

TABLE III
DECODING ACCURACY FOR SUBJECTS WITH THE 25% HIGHEST AND 25% LOWEST HO ACCURACY FOR THE OPENBMI DATASET

| Subjects | Methods | OpenBMI (HO) |
|---|---|---|
| Top 25% | FBCSP-SVM | $83.04 \pm 11.34$ |
| | Shallow ConvNet | $84.92 \pm 9.28$ |
| | Deep ConvNet | $77.19 \pm 8.73$ |
| | EEGNet | $79.35 \pm 7.32$ |
| | FBCNet | $88.30 \pm 8.12$ |
| | FBMSNet | $\mathbf{91.84 \pm 5.79}$ |
| Bottom 25% | FBCSP-SVM | $47.69 \pm 1.93$ |
| | Shallow ConvNet | $46.38 \pm 3.51$ |
| | Deep ConvNet | $49.46 \pm 2.27$ |
| | EEGNet | $51.38 \pm 2.14$ |
| | FBCNet | $51.50 \pm 1.83$ |
| | FBMSNet | $\mathbf{53.35 \pm 2.19}$ |

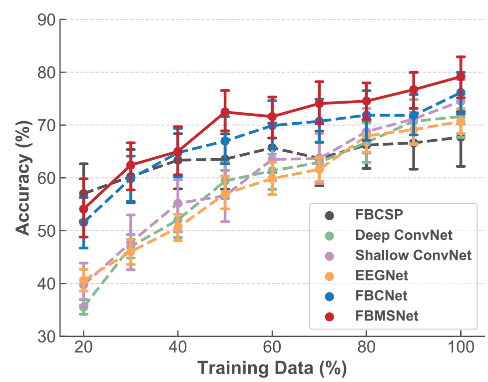


Fig. 3. Sensitivity of the MI decoding algorithms with various amounts of training data, where the values on the x-axis represent the proportion of the training set used for training. The results are shown as the mean ± SEM (standard error of the mean (SEM)) of subjects.

the performance of FBMSNet and the compared methods with various amounts of training data. The sensitivity of the classification algorithm to fewer training samples was evaluated with the BCIC-IV-2a dataset under HO analysis. Specifically, all methods were trained using various fractions of the training dataset ranging from 20% to 100% in steps of 10%. Trained models were then tested on separate session 2 data, and the test accuracy was analyzed as a function of training data percentage.

As shown in Fig. 3, EEGNet and Deep ConvNet were highly sensitive to small training sets, with an accuracy drop of approximately 35% when the number of training trials was reduced to 20%. In contrast, FBCSP-SVM was least affected by a reduction in the training set. In fact, the drop in accuracy using FBCSP-SVM only became statistically significant when the portion of training samples was reduced to 30%. Furthermore, compared to Deep ConvNet and EEGNet, FBCSP-SVM achieved significantly larger classification accuracy when the amount of training data was reduced below 50%. However, the maximum accuracy achieved by FBCSP-SVM, when using the complete training data was much lower compared to EEGNet and Deep ConvNet (see Table II). Additionally, the accuracy curve of FBMSNet closely followed the characteristics of FBCSP-SVM with a similar sensitivity to the reduced training set as that of FBCSP-SVM. A similar trend can also be found for FBCNet. However, FBMSNet outperformed FBCNet when more than 50% of the training data were available, and the difference was statistically significant ($p < 0.05$ for BCIC-IV-2a data in HO analysis).

In summary, Deep ConvNet and EEGNet are highly sensitive to small training sets, whereas FBCSP-SVM is relatively less sensitive. FBMSNet matches the accuracy achieved by deep learning methods in the presence of ample training data while retaining better performance even when the training set is small.

*Result III: FBMSNet achieved minimal overlap of embedded features between different classes*

To compare the abilities of deep learning methods in the extraction of highly discriminative features from EEG signals, we used the t-SNE method [28] implemented through scikit-learn [29] to visualize the generalized features learned by different deep learning methods in the two-dimensional embedding space.

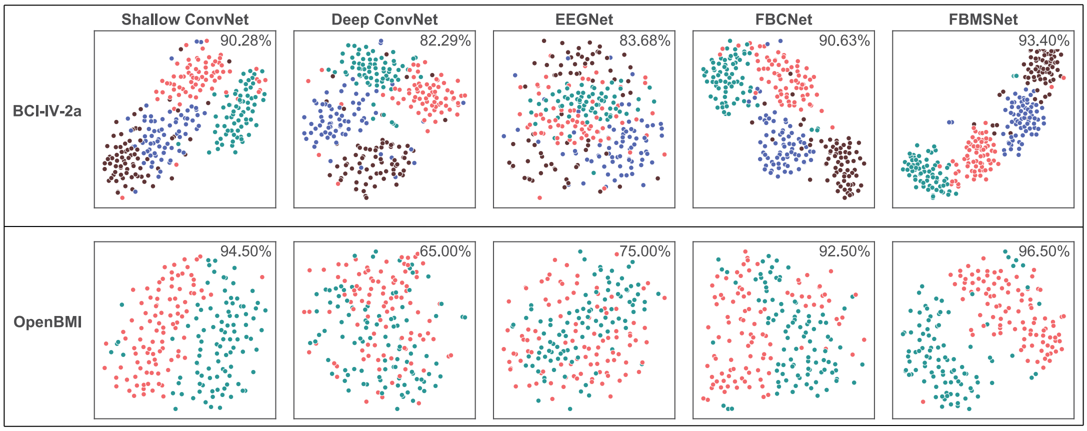


Fig. 4. Visualization of features inputted into the classification layer and corresponding raw data using two-dimensional t-SNE projection. The top row is the visualization from subject A03 of BCI-IV-2a dataset, and the bottom row is the result from subject s3 of OpenBMI dataset. Red, green, blue and brown points represent the MI of left hand, right hand, foot and tongue, respectively.

Fig. 4 shows the t-SNE visualization results for the 2D embedding features of the high-dimensional features at the input of the final classification layer for Shallow ConvNet, Deep ConvNet, EEGNet-8,2, FBCNet and FBMSNet for two subjects. There is a large amount of overlap between the different classes in the raw data. For all deep learning methods, there are different degrees of overlap between the embedding features of MI signals from different classes, but FBMSNet resulted in a minimal degree of overlap. In particular, from the results of FBCNet and FBMSNet, a significant portion of the previously intertwined samples have been separated, except for a small number of samples that are still mixed with the dissidents. Additionally, as shown in the example on the OpenBMI dataset, the features from FBMSNet are presented to be more separable than those generated by FBCNet. This provides evidence that our FBMSNet method produces superior performance owing to the greater quality representation of the MI in the learned latent features.

## V. Discussion

### A. Effects of Mixed Convolution

Due to the high temporal resolution, EEG data contain abundant temporal information. Several studies have shown the effect of multiscale feature extraction at time domain. To analyze the role of the mixed-scale temporal layers of FBMSNet, we performed an ablation experiment removing the central loss. In this experiment, the main difference between FBMSNet and FBCNet is the existence of mixed convolution. We tested FBMSNet without center loss against FBCNet in the HO analysis of the OpenBMI dataset. As shown in Fig. 5, for all 54 subjects, FBMSNet (without center loss) provided significantly larger accuracy than FBCNet ($p < 0.05$). As suggested in [30], a BCI system with $>70\%$ binary classification accuracy is generally considered to be usable by healthy subjects and stroke patients. We further compared the performance of FBMSNet (without center loss) for subjects with FBCNet accuracy $>70\%$ and $<70\%$. As shown in Fig. 5, the superior performance of FBMSNet is more obvious for the subjects whose decoding accuracy is less than 70% by FBCNet. This comparison demonstrates the positive effect of multiscale temporal feature extraction on MI classification.

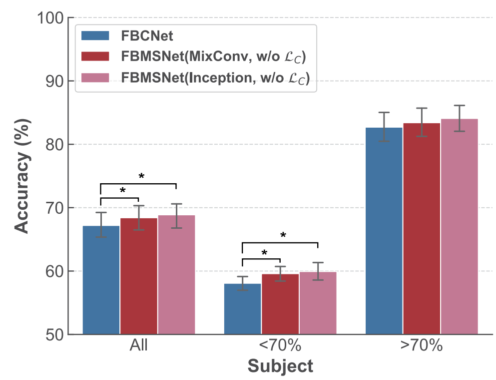


Fig. 5. Comparisons of FBCNet, FBMSNet (MixConv) without center loss, and FBMSNet (Inception) without center loss in the HO analysis of OpenBMI dataset. Results are shown as the mean $\pm$ SEM (standard error of the mean (SEM)) of subjects. The * indicates $p < 0.05$.

Certainly, there exist other choices to extract multiscale temporal features, such as the inception module [31], which has been recently employed for temporal sequence classification [32] and BCI [13], [24]. Here, we compared the performance of the MixConv and Inception modules in FBMSNet. According to the HO analysis results of the OpenBMI dataset (see Fig. 5), both the MixConv and Inception modules provide significantly higher accuracy than FBCNet for all subjects and the subjects whose accuracy using FBCNet is $<70\%$, but the difference between the MixConv and Inception modules is marginal and not statistically significant. However, the number of trainable parameters of the Inception module ($1.89 \times 10^4$) is much larger than that of the MixConv module ($4.4 \times 10^3$). According to our numerical experiments, MixConv requires 56.88% less time to train for 10 epochs compared to Inception. Hence, considering the implementation of future mobile BCI applications, in this work, we choose the MixConv module to extract multiscale temporal information of EEG MI data.

TABLE IV
DECODING ACCURACY OF FBMSNET WITH AND WITHOUT CENTER LOSS FOR OPENBMI DATASET IN HO ANALYSIS

| Subject | Model | OpenBMI (HO) |
|---|---|---|
| All | FBMSNet(w/o $\mathcal{L}_C$) | $68.41 \pm 14.16$ * |
| | FBMSNet(w/ $\mathcal{L}_C$) | $70.05 \pm 14.51$ |
| Acc.<70% | FBMSNet(w/o $\mathcal{L}_C$) | $58.91 \pm 6.40$ * |
| | FBMSNet(w/ $\mathcal{L}_C$) | $61.38 \pm 7.94$ |
| Acc.>70% | FBMSNet(w/o $\mathcal{L}_C$) | $83.33 \pm 9.32$ |
| | FBMSNet(w/ $\mathcal{L}_C$) | $83.66 \pm 11.79$ |

*indicates $p < 0.05$.

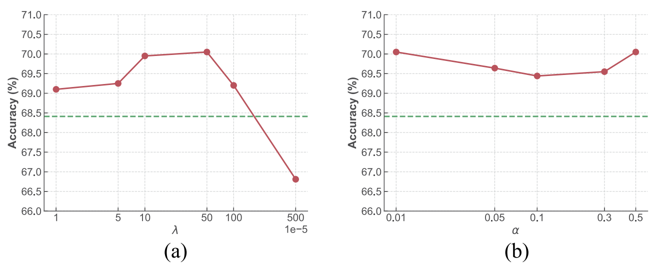

Fig. 6. Sensitivity of the classification algorithm to $\lambda$ and $\alpha$. Results are shown for the OpenBMI dataset in HO schema. (a) Result with different $\lambda$ and fixed $\alpha = 0.01$; (b) Result with different $\alpha$ and fixed $\lambda = 0.0005$, where the green dashed line represents the accuracy by FBMSNet with $\lambda = 0$.

## B. Effects of Center Loss

Whereas cross-entropy loss aims to minimize misclassification of subject movement intent, center loss minimizes the sum of the embedded space distance of samples in a class to its center, making the samples belonging to the same class compact in the feature space. However how does the center loss affect the MI decoding accuracy for different subjects? To test the influence of center loss, we compared the performance of FBMSNet with and without center loss for subjects with FBMSNet accuracy >70% and <70%.

As shown in Table IV, for all 54 subjects in the OpenBMI dataset, the center loss resulted in an average accuracy improvement of 1.64%. For subjects with an FBMSNet accuracy <70%, the center loss has a significant positive impact on the classification performance ($p = 0.015$). However, for subjects with FBMSNet accuracy >70%, the center loss has a marginal impact on the decoding performance. This experimental result suggests that bringing together samples belonging to the same class using center loss can improve the MI decoding performance, especially for subjects who are inexperienced with MI.

## C. Influence of the Values of $\lambda$ and $\alpha$

FBMSNet employs a combination of cross-entropy cost and center loss as the final cost function, as shown in Eq. (3). To implement the center loss, two hyperparameters, i.e., the learning rate of feature centers $\alpha$ and the trade-off value between the cross-entropy and center loss, are needed. To test the influence of the two hyperparameters, we conducted a comparison of FBMSNet with difference values of $\lambda$ and $\alpha$ in the HO analysis of the OpenBMI dataset.

As shown in Fig. 6(a), when $\lambda$ increases from 0 to 0.001, FBMSNet shows increased MI decoding accuracy. Moreover, features are discriminative when $\lambda \in [0.0001, 0.0005]$. If we continue to increase the value of $\lambda$, FBMSNet depicts decreased MI decoding performance. However, the learning rate of feature centers, $\alpha$, has little influence on the accuracy of MI decoding (see Fig. 6(b)), which is in line with the result in [22]. Hence, in this work, we experimentally set the values of $\lambda$ and $\alpha$ to 0.0005 and 0.01, respectively, obtaining higher decoding performance when no center loss is considered (e.g., $\lambda = 0$).

## D. Activation Function in the Spatial Convolution Block

Since FBMSNet is designed based on FBCNet, which employs the Swish activation function in the spatial convolution block, for fair comparison, in this work, we employed the same activation function as FBCNet. We have also applied FBMSNet to the hold-out analysis on the BCIC-IV-2a dataset with the Swish and ReLU activation functions respectively. FBMSNet with Swish slightly outperforms (79.17%) that with the ReLU activation function (78.32%), but this improvement is not significant (Kruskal–Wallis test, $p$-value $= 0.3105$).

## E. Limitations and Future Work

In this work, we only test the performance of the proposed method with the MI of different limbs, such as the left hand, right hand, both feet and tongue. To achieve a more versatile and practical SMR-based BCI, neural decoding of highly dexterous movements is a crucial element. Our future work will apply FBMSNet to decode the movement of localized arm parts, such as hand grasping and wrist twisting [33] or MI of different directions [34]. Additionally, the generalization ability of FBMSNet cross subjects will be explored in our future work.

One advantage of deep neural networks is that they can absorb very large amounts of data to learn excellent feature representations, which in turn allows for better performance. Due to the limited amount of training data in BCI, recent studies have proposed data augmentation methods to further improve the performance of deep neural networks [11], [13]. Additionally, due to the lack of subject-specific data, many studies have employed the adaptive transfer learning to fine-tune the model for the target subject [35]. In future work, we will combine these data augmentation methods and transfer learning strategy to improve the decoding performance of FBMSNet. Moreover, as suggested by [14], [36], EEG representation including the information of time, frequency and location of electrodes is able to enhance the MI decoding accuracy. These EEG representations may be employed as the network input, such as the 3D representation in [12].

# VI. CONCLUSION

In this paper, we propose a CNN, termed FBMSNet, which employs the mixed depthwise convolution to efficiently extract multiscale temporal features for EEG MI recognition. To learn more discriminative features, we combined the center loss with the CE loss. Validations on two publicly available benchmark datasets demonstrate the superiority of the proposed method. The code of FBMSNet can be openly accessed. The proposed method yields promising MI decoding performance. In the future, we will also apply the proposed MI decoding model in areas such as rehabilitation engineering and brain function analysis.

## References


[1] D. Jean, "The neurophysiological basis of motor imagery," *Behav. Brain Res.*, vol. 77, no. 1, pp. 45–52, 1996.

[2] M. Rashid et al., "Current status, challenges, and possible solutions of EEG-based brain-computer interface: A comprehensive review," *Front. Neurorobot.*, vol. 14, 2020, Art. no. 25.

[3] F. Pichiorri et al., "Brain–computer interface boosts motor imagery practice during stroke recovery," *Ann. Neurol.*, vol. 77, no. 5, pp. 851–865, 2015.

[4] K. K. Ang and C. Guan, "Brain–computer interface for neurorehabilitation of upper limb after stroke," *Proc. IEEE Proc. IRE*, vol. 103, no. 6, pp. 944–953, Jun. 2015.

[5] R. Bousseta et al., "EEG based brain computer interface for controlling a robot arm movement through thought," *IRBM*, vol. 39, no. 2, pp. 129–135, Apr. 2018.

[6] B. J. Edelman et al., "Noninvasive neuroimaging enhances continuous neural tracking for robotic device control," *Sci. Robot.*, vol. 4, no. 31, 2019, Art. no. 6844.

[7] J. Tang et al., "Towards BCI-actuated smart wheelchair system," *Biomed. Eng. Online*, vol. 17, no. 1, pp. 1–22, 2018.

[8] K. K. Ang et al., "Filter bank common spatial pattern (FBCSP) in brain-computer interface," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, 2008, pp. 2390–2397.

[9] R. T. Schirrmeister et al., "Deep learning with convolutional neural networks for EEG decoding and visualization," *Hum. Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

[10] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, 2018, Art. no. 056013.

[11] G. Dai et al., "HS-CNN: A CNN with hybrid convolution scale for EEG motor imagery classification," *J. Neural Eng.*, vol. 17, no. 1, 2020, Art. no. 016025.

[12] X. Zhao, H. Zhang, G. Zhu, F. You, S. Kuang, and L. Sun, "A multi-branch 3D convolutional neural network for EEG-based motor imagery classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 10, pp. 2164–2177, Oct. 2019.

[13] C. Zhang, Y. Kim, and A. Eskandarian, "EEG-inception: An accurate and robust end-to-end neural network for EEG-based motor imagery classification," *J. Neural Eng.*, vol. 18, no. 4, 2021, Art. no. 046014.

[14] N. Robinson, S. Lee, and C. Guan, "EEG representation in deep convolutional neural networks for classification of motor imagery," in *Proc. IEEE Int. Conf. Syst., Man Cybern.*, 2019, pp. 1322–1326.

[15] R. Mane et al., "FBCNet: A multi-view convolutional neural network for brain-computer interface," 2021, *arXiv:2104.01233*.

[16] W. Liu et al., "Large-margin softmax loss for convolutional neural networks," in *Proc. Int. 33rd Conf. Mach. Learn.*, 2016, pp. 507–516.

[17] H. Wang et al., "CosFace: Large margin cosine loss for deep face recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 5265–5274.

[18] L. Yang et al., "Motor imagery EEG decoding method based on a discriminative feature learning strategy," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 368–379, 2021.

[19] S. Pradeep et al., "Towards adaptive classification for BCI," *J. Neural Eng.*, vol. 3, no. 1, pp. R13–R23, 2006.

[20] A. Satti et al., "A covariate shift minimisation method to alleviate non-stationarity effects for an adaptive brain-computer interface," in *Proc. 20th Int. Conf. Pattern Recognit.*, 2010, pp. 105–108.

[21] P. Autthasan et al., "MIN2Net: End-to-end multi-task learning for subject-independent motor imagery EEG classification," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 6, pp. 2105–2118, Jun. 2021.

[22] Y. Wen et al., "A discriminative feature learning approach for deep face recognition," in *Proc, Eur. Conf. Comput. Vis.*, 2016, pp. 499–515.

[23] T. Mingxing and V. L. Quoc, "MixConv: Mixed depthwise convolutional Kernels," in *Proc. 30th Brit. Mach. Vis. Conf.*, Cardiff, U.K., Sep. 9–12, 2019.

[24] Y. Ding et al., "TSception: A deep learning framework for emotion detection using EEG," in *Proc. Int. Joint Conf. Neural Netw.*, 2020, pp. 1–7.

[25] M. Tangermann et al., "Review of the BCI competition IV," *Front. Neurosci.*, vol. 6, 2012, Art. no. 55.

[26] M. Lee et al., "EEG dataset and OpenBMI toolbox for three BCI paradigms: An investigation into BCI illiteracy," *GigaScience*, vol. 8, no. 5, 2019, Art. no. giz002.

[27] D. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Representations*, 2015.

[28] V. der Maaten and L. G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

[29] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[30] X. Shu et al., "Fast recognition of BCI-inefficient users using physiological features from EEG signals: A screening study of stroke patients," *Front. Neurosci.*, vol. 12, 2018, Art. no. 93.

[31] C. Szegedy et al., "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2818–2826.

[32] I. Fawaz et al., "Inceptiontime: Finding alexnet for time series classification," *Data Mining Knowl. Discov.*, vol. 34, no. 6, pp. 1936–1962, Nov. 2020.

[33] J. Jeong et al., "Multimodal signal dataset for 11 intuitive movement tasks from single upper extremity during multiple recording sessions," *GigaScience*, vol. 9, no. 10, pp. 1–15, Oct. 2020.

[34] V. S. Handiru, A. P. Vinod, and C. Guan, "EEG source space analysis of the supervised factor analytic approach for the classification of multi-directional arm movement," *J. Neural Eng.*, vol. 14, no. 4, 2017, Art. no. 046008.

[35] K. Zhang et al., "Adaptive transfer learning for EEG motor imagery classification with deep convolutional neural network," *Neural Netw.*, vol. 136, pp. 1–10, 2021.

[36] M. A. Li, J. F. Han, and L. J. Duan, "A novel MI-EEG imaging with the location information of electrodes," *IEEE Access*, vol. 8, pp. 3197–3211, 2019.